Conditional Expectations, Beck Defects, and the Shadow Algebra of the Posterior Tower

Revised version: the corrected interchange theorem, canonical shadows, and the algebra and topology of how the tower acts

Vinay K. Awasthi
Department of Applied Mathematics and Statistics, Johns Hopkins University

October 13, 2026

Abstract

The three-level posterior tower [1] has two candidate mixed distributive laws, and both fail some Beck axioms. An earlier version of this paper proposed to organise the failures by two "shadows", left and right pooling operators, and asserted an interchange law for them. That version had three defects. First, its interchange argument used the identity $P_B P_A = P_{A \vee B}$, which is false. Second, it defined the shadows by assigning each point to its nearest atom, and composites of such operators are not composites of conditional expectations. Third, it built a double category whose squares were not typed. This revision replaces all three.

Algebra. For finite partitions A, B of a full-support probability space, five conditions are equivalent: $E_A E_B = E_B E_A$; $E_A E_B = E_{A \wedge B}$; conditional independence of A and B given the meet $A \wedge B$; a rank-one contingency block on every connected component of the bipartite cell graph; and singular values of the normalised contingency matrix lying in $\{0, 1\}$. The join identity holds only when $A = B$. In general, $(E_A E_B)^n$ converges to $E_{A \wedge B}$ at the exact rate c^{2n-1}, where c is the Friedrichs cosine. This gives a canonical *lax interchange*: the two composites differ, but both lie above the meet in convex order. Conditional independence given a third variable is not enough, and the earlier Gaussian interchange proposition is withdrawn.

Canonical shadows. A coherent context determines a finite probability space shaped as a star, with centre the level-one predictive and leaves the level-two law, the hypothesis label and the observation. Shadows are conditional expectations on this fixed space, each conditioning on the centre together with a set of leaves. They form a commutative band isomorphic to a quotient of the Boolean lattice of leaf sets. The interchange law therefore holds strictly for the canonical shadows, and it fails, in a way the spectral criterion detects, when the centre is dropped.

Topology and order. The barycentre map is a 1-Lipschitz continuous retraction with two continuous splittings, $e_\perp = \eta\mu$ and $e_\top = \mathsf{D}(\eta)\mu$. These form a left-zero band, $e_\top$ is affine, $e_\perp$ is not, and they pick out the bottom and top of every barycentre fibre in convex order. A coherent context is a three-step martingale in the simplex. The shift law's level-one unit defect is the full convex-order diameter of a fibre, and its multiplication defect is one conditional-expectation step. Nearest-atom shadows are discontinuous across Voronoi walls; the canonical ones are continuous.

The analytic sections are corrected as well: the Kleisli-to-convex functor is fully faithful, the non-Archimedean tensor construction is ill-defined as stated, and second-order Gaussian moments cannot see mixtures. Every finite claim is checked in exact arithmetic (`check_shadows.py`, 26 checks), together with an audit of the five scripts that accompanied the earlier version. That audit explains why `check_span.py` could not separate the two shadows.

Contents

1 Introduction **3**
- 1.1 The problem . . . 3
- 1.2 What this revision corrects . . . 3
- 1.3 Main results . . . 3
- 1.4 Conventions . . . 3

2 The tower recalled **4**

3 Algebra I: conditional expectations on a finite space **4**
- 3.1 The partition lattice . . . 4
- 3.2 The commutation theorem . . . 4
- 3.3 The general case: rate and lax interchange . . . 6
- 3.4 The Gaussian correction . . . 7

4 Canonical shadows **7**
- 4.1 Why label-defined measure-valued shadows do not compose . . . 7
- 4.2 Labelled random measures . . . 8
- 4.3 The star space of a context . . . 8

5 Algebra II and topology: how the tower acts **10**
- 5.1 Two idempotents . . . 10
- 5.2 Barycentre fibres . . . 10
- 5.3 A coherent context is a martingale . . . 11
- 5.4 The Beck defects located . . . 11
- 5.5 Continuity: canonical versus label-defined shadows . . . 11

6 Two-dimensional structure: what is and is not a double category **12**
- 6.1 Why the previous double-category theorem is withdrawn . . . 12
- 6.2 Quintets of a locally preordered 2-category . . . 12
- 6.3 What the tower supplies, and what it lacks . . . 12
- 6.4 Options A and B . . . 13

7 The Giry extension **13**

8 Analytic and algebraic outlook, corrected **14**
- 8.1 The finite stochastic-to-convex functor . . . 14
- 8.2 The Berkovich construction . . . 14
- 8.3 Moment varieties . . . 14

9 Computational checks and the audit of the earlier scripts **15**
- 9.1 The new checks . . . 15
- 9.2 Audit of the five scripts of the previous version . . . 15

10 Remark added to the tower paper **16**

11 Summary and open problems **16**

1 Introduction

1.1 The problem

A three-level posterior tower records three things: a predictive $\rho \in \mathsf{D}X$, a law $\Pi \in \mathsf{D}^2X$ over predictives, and a law $\Omega \in \mathsf{D}^3X$ over such laws. It also carries a shadow $\Sigma \in \mathsf{Sh}(X) = \mathsf{D}(\Theta \times \mathsf{D}X)$ of labelled hypotheses. The absolute tower $\mathsf{C}_3X = X \times \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{Sh}X$ is a comonad. Two candidate mixed laws $\mathsf{DC}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ fail Beck axioms [1, Thms. 8.3, 8.6, 8.7]. The shift law λ^σ satisfies (Co) and fails (U), (M) and (Cm). The averaging law $\bar{\lambda}$ satisfies (U), (Co) and (M) and fails (Cm). The motivating question of this paper is whether two ways of pooling information, called the left and right shadows, organise these failures. A natural sub-question is whether the two shadows commute.

1.2 What this revision corrects

Table 1 lists every claim of the previous version that changes. The guiding principle: *operators are composed on a fixed probability space.* Composing measure-valued maps that recompute their partitions after each step is a different operation, and most of the earlier difficulties come from mixing the two.

1.3 Main results

(A) **Commutation theorem** (theorem 3.3). For finite partitions, five equivalent conditions characterise commuting conditional expectations. One of them is spectral and one is combinatorial: rank-one blocks on the connected components of a bipartite graph. The first three are equivalent on every probability space.

(B) **Rate and lax interchange** (theorem 3.6 and corollary 3.8). $\|(E_AE_B)^n - E_{A\wedge B}\| = c^{2n-1}$, and both composites are bounded below by $E_{A\wedge B}$ in convex order. This settles the finite general case.

(C) **Shadow algebra** (theorem 4.8). The canonical shadows of a context form a commutative band, a homomorphic image of the Boolean lattice of leaf sets.

(D) **Two idempotents and the fibre** (propositions 5.1 and 5.2). $e_\perp$ and $e_\top$ are continuous, form a left-zero band, and bound every barycentre fibre in convex order.

(E) **The tower as a martingale** (theorem 5.3). The levels of a coherent context are the successive positions of one simplex-valued martingale.

(F) **Beck defects located** (proposition 5.4). The level-one unit defect of λ^σ is the pair $(e_\perp\Pi, e_\top\Pi)$, and its multiplication defect is one conditional-expectation step on the sample index.

1.4 Conventions

Throughout, X is a finite set and D is the finite distribution monad, with unit $\eta_X(x) = \delta_x$ and multiplication $\mu_X(\Pi) = \sum_\rho \Pi(\rho)\rho$. We identify $\mathsf{D}X$ with the simplex $\Delta_X \subset \mathbb{R}^X$, with the ℓ^1 metric. A *finite probability space* (I, p) has $p(i) > 0$ for all i. On such a space, σ-algebras are partitions. For a partition A write E_A for the conditional expectation, acting coordinatewise on $\mathbb{R}^X$-valued functions: $(E_AZ)(i) = \sum_{j\in A(i)} p(j)Z(j)/p(A(i))$, where $A(i)$ is the cell of i.

2 The tower recalled

We use the definitions of [1] without change; we recall only what is needed. A hypothesis is a pair $h = (\theta, r) \in \mathsf{Hyp}X = \Theta \times \mathsf{D}X$, with predictive $q(h) = r$. A context is $c = (\rho, \Pi, \Omega, \Sigma)$ with $\rho \in \mathsf{D}X$, $\Pi \in \mathsf{D}^2X$, $\Omega \in \mathsf{D}^3X$ and $\Sigma \in \mathsf{D}(\mathsf{Hyp}X)$. It is *coherent* if

$$\mu(\Pi) = \rho, \qquad \mu(\Omega) = \Pi, \qquad \mathsf{D}(q)(\Sigma) = \Pi. \tag{1}$$

The shift law λ^σ transports a sample $M = \sum_i a_i \delta_{(x_i, c_i)}$ by keeping each sample's context entries, and $\bar{\lambda}$ transports it by averaging contexts. Their Beck properties were listed above. At level one, [1, Prop. 13.8] computes the two sides of (U) and (M) for λ^σ. The tower paper also proves [1, Prop. 13.9, Thm 13.15, Prop. 13.17] that every order which reads pooling monotonically orients the two defects oppositely, and [1, Thm 13.16] that the (Cm) defect of $\bar{\lambda}$ is label pooling at every level. That paper never composes conditional expectations onto two different σ-algebras as if their product were a conditional expectation: its pooling relations are taken on the sample index, and its order is the transitive closure. The correction of this paper therefore does not propagate into the tower paper; section 10 records the one clarifying remark added there.

3 Algebra I: conditional expectations on a finite space

3.1 The partition lattice

Partitions of I form a lattice under refinement. The *join* $A \vee B$ is the common refinement, with cells $a \cap b$. The *meet* $A \wedge B$ is the finest common coarsening. As σ-algebras, $A \vee B = \sigma(A \cup B)$ and $A \wedge B = A \cap B$.

Lemma 3.1 (The meet is π_0 of a bipartite graph)**.** *Let $G(A, B)$ be the bipartite graph whose vertices are the cells of A and of B, with an edge a–b whenever $a \cap b \neq \emptyset$. The cells of $A \wedge B$ are the unions of the cells in the connected components of $G(A, B)$. Moreover, $L^2(A) \cap L^2(B) = L^2(A \wedge B)$.*

Proof. A set lies in $A \cap B$ iff it is a union of A-cells and a union of B-cells. Such a set is closed under the edge relation, so it is a union of components, and every union of components qualifies. A function lies in $L^2(A) \cap L^2(B)$ iff it is constant on A-cells and on B-cells, that is, iff it is constant on components. □

So the meet is topological: it is the set of path components of the cell graph. The join is order-theoretic. The interchange question concerns the meet.

3.2 The commutation theorem

Write $N_{ab} = p(a \cap b)$ for the contingency matrix, and $K = D_A^{-1/2} N D_B^{-1/2}$ with $D_A = \mathrm{diag}(p(a))$, $D_B = \mathrm{diag}(p(b))$.

Lemma 3.2 (A sufficient pair)**.** *On any probability space, let $C \subseteq A \cap B$ be sub-σ-algebras such that A and B are conditionally independent given C. Then $E_A E_B = E_C = E_B E_A$, and $C = A \cap B$ up to null sets.*

Proof. Let $V \in L^2(B)$ and $U \in L^2(A)$. By conditional independence and $C \subseteq A$, $\mathbb{E}[UV] = \mathbb{E}[\mathbb{E}[U \mid C]\mathbb{E}[V \mid C]] = \mathbb{E}[U\, \mathbb{E}[V \mid C]]$. So $E_A V = E_C V$, that is, $E_A E_B = E_C E_B = E_C$; the last step holds because $C \subseteq B$. The other order is symmetric. The range of E_C is $L^2(C)$, and it equals the range $L^2(A) \cap L^2(B) = L^2(A \cap B)$ of the commuting product. □

Theorem 3.3 (Commutation)**.** *For partitions A, B of a finite probability space the following are equivalent:*

(i) $E_A E_B = E_B E_A$*;*

(ii) $E_A E_B = E_{A \wedge B}$*;*

(iii) *A and B are conditionally independent given $A \wedge B$;*

(iv) *on each connected component γ of $G(A, B)$, the block $N|_\gamma$ has rank one: $p(a \cap b)\, p(\gamma) = p(a)\, p(b)$ for all cells $a, b \subseteq \gamma$;*

(v) *every singular value of K is 0 or 1.*

Here (i)⇔(ii)⇔(iii) hold on every probability space (with $A \wedge B$ read as $A \cap B$). The value 1 in (v) occurs with multiplicity equal to the number of components of $G(A, B)$. Nestedness ($A \subseteq B$ or $B \subseteq A$) implies (iv).

Proof. (iii)⇒(ii) is lemma 3.2 with $C = A \wedge B$. (ii)⇒(i): take adjoints. $E_A E_B = E_{A \wedge B}$ is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_A E_B$.

(i)⇒(ii): a product of commuting orthogonal projections is the orthogonal projection onto the intersection of their ranges, which is $L^2(A \wedge B)$ by lemma 3.1.

(ii)⇒(iii): for $U \in L^2(A)$ and $V \in L^2(B)$ put $C = A \wedge B$. Then $\mathbb{E}[UV \mid C] = \mathbb{E}[U\, E_A V \mid C] = \mathbb{E}[U\, E_C V \mid C] = E_C U \cdot E_C V$, since $E_A V = E_A E_B V = E_C V$ is C-measurable.

(iii)⇔(iv): on a finite space it suffices to test indicators. Cells of A and B lie inside single components, so conditional independence given C is the identity $p(a \cap b \mid \gamma) = p(a \mid \gamma) p(b \mid \gamma)$ for $a, b \subseteq \gamma$. This says that $N|_\gamma$ is the outer product of its margins divided by $p(\gamma)$, which is the same as rank one, because a rank-one nonnegative matrix equals the outer product of its margins over its total.

(ii)⇔(v): in the orthonormal bases $1_a/\sqrt{p(a)}$ of $L^2(A)$ and $1_b/\sqrt{p(b)}$ of $L^2(B)$, the operator $E_A|_{L^2(B)}$ has matrix K. Its singular values are the cosines of the principal angles between $L^2(A)$ and $L^2(B)$. The value 1 occurs exactly on $L^2(A) \cap L^2(B) = L^2(A \wedge B)$, whose dimension is the number of components. The equality $E_A E_B = E_{A \wedge B}$ holds iff $L^2(A) \ominus L^2(A \wedge B) \perp L^2(B) \ominus L^2(A \wedge B)$, that is, iff every other principal angle is $\pi/2$. If $A \subseteq B$ then each A-cell is a component and (iv) is trivial. □

The combinatorial and spectral conditions (iv) and (v) make the theorem checkable. In exact arithmetic, check A1 confirms the five-way equivalence on 600 random instances: 449 commuting (including constructed conditionally independent and nested instances) and 151 non-commuting.

Proposition 3.4 (The join identity)**.** *$E_A E_B = E_{A \vee B}$ holds iff $A = B$ (up to null sets). In particular the identity $P_B P_A = P_{A \vee B}$ used in the previous version fails for every pair of distinct σ-algebras.*

Proof. The range of $E_A E_B$ lies in $L^2(A)$. If it equals $L^2(A \vee B) \supseteq L^2(B)$, then $B \subseteq A$, so $E_A E_B = E_B$ and $E_{A \vee B} = E_A$, whence $E_A = E_B$ and $A = B$. The converse is trivial. □

Example 3.5 (Non-commuting pair)**.** Let $I = \{1, 2, 3, 4\}$ with $p = (\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$, $Z(i) = i$, $A_L = \{\{1, 2\}, \{3, 4\}\}$ and $A_R = \{\{1, 3\}, \{2, 4\}\}$. Then $(E_{A_R} E_{A_L} Z)(1) = \frac{15}{8}$ and $(E_{A_L} E_{A_R} Z)(1) = 2$. The graph $G(A_L, A_R)$ is a 4-cycle, so the meet is trivial. The block $N = \left(\begin{smallmatrix} 3/8 & 1/8 \\ 1/8 & 3/8 \end{smallmatrix}\right)$ has rank two, and K has singular values 1 and $\frac{1}{2}$. So $c = \frac{1}{2}$ and $\|(E_{A_L} E_{A_R})^n - E_{\text{triv}}\| = 2^{1-2n}$ (check A3).

3.3 The general case: rate and lax interchange

When (i)–(v) fail, the two composites differ. Neither is a conditional expectation (a product of two orthogonal projections is a projection only when they commute), but both are controlled by the meet.

Theorem 3.6 (Alternating projections, exact rate)**.** *Let c be the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. In the finite case c is the largest singular value of K strictly below 1, and $c = 0$ if there is none. Then for all $n \geq 1$,*

$$\|(E_A E_B)^n - E_{A\wedge B}\| = c^{2n-1}$$

in operator norm on $L^2(p)$. On a general probability space the same identity holds with c the Friedrichs cosine; then $c < 1$ iff $L^2(A)+L^2(B)$ is closed, and otherwise the convergence $(E_A E_B)^n \to E_{A\wedge B}$ holds only strongly and can be arbitrarily slow.

Proof. This is the theorem of Kayalar and Weinert [8] for two orthogonal projections, applied with $L^2(A) \cap L^2(B) = L^2(A \wedge B)$ (lemma 3.1). Strong convergence is von Neumann's theorem [12], and arbitrarily slow convergence when $c = 1$ is in [4]. The finite identification of c follows from the proof of theorem 3.3(v). □

Check A4 verifies c^{2n-1} to 10^{-9} for $n = 1, \ldots, 6$ on 157 non-commuting pairs.

Definition 3.7 (Convex order)**.** For finitely supported laws Q, Q' on a convex set K, $Q \leq_{\mathrm{cx}} Q'$ if $\int \varphi\, dQ \leq \int \varphi\, dQ'$ for all convex φ. By Strassen's theorem [11], this holds iff $Q = \mathrm{Law}(\mathbb{E}[U' \mid G])$ for some $U' \sim Q'$ on a finite space and some sub-σ-algebra G.

Corollary 3.8 (Lax interchange)**.** *For any $\mathbb{R}^X$-valued Z on (I, p),*

$$E_{A\wedge B} E_A E_B = E_{A\wedge B} = E_{A\wedge B} E_B E_A,$$

and hence

$$\mathrm{Law}(E_{A\wedge B} Z) \;\leq_{\mathrm{cx}}\; \mathrm{Law}(E_A E_B Z) \;\leq_{\mathrm{cx}}\; \mathrm{Law}(E_B Z) \;\leq_{\mathrm{cx}}\; \mathrm{Law}(Z),$$

together with the symmetric chain through $E_B E_A Z$. The sequence $\mathrm{Law}((E_A E_B)^n Z)$ decreases in convex order, and

$$W_2\big(\mathrm{Law}((E_A E_B)^n Z), \mathrm{Law}(E_{A\wedge B} Z)\big) \leq c^{2n-1}\, \|Z - E_{A\wedge B} Z\|_{L^2(p)}.$$

When the conditions of theorem 3.3 fail, the two composites differ as operators, and each strictly exceeds the meet in L^2 energy on the vectors where they differ from $E_{A\wedge B} Z$.

Proof. $A \wedge B \subseteq A, B$, so by the tower property $E_{A\wedge B} E_A = E_{A\wedge B} = E_{A\wedge B} E_B$. Each displayed law is the law of a conditional expectation of the next on the same space, hence $\leq_{\mathrm{cx}}$ it by Strassen. For the rate, note $(E_A E_B)^n - E_{A\wedge B} = ((E_A E_B)^n - E_{A\wedge B})(I - E_{A\wedge B})$, because both $(E_A E_B)^n E_{A\wedge B}$ and $E_{A\wedge B}^2$ equal $E_{A\wedge B}$. Then apply theorem 3.6, using that the identity coupling on I bounds W_2. Strictness follows from Pythagoras for orthogonal projections. □

So in the general (non-nested, non-conditionally-independent) case the interchange law does not hold strictly. It holds *laxly*: the two composites form a cospan

$$E_A E_B \implies E_{A\wedge B} \impliedby E_B E_A$$

of convex-order comparisons into the meet. These comparisons are identities exactly under theorem 3.3, and their size decays at the rate of theorem 3.6. In finite dimensions this answers the "general case" question. In infinite dimensions c can equal 1 (section 7).

3.4 The Gaussian correction

The previous version (its Prop. 11.7) argued as follows: the two channels X_L, X_R are conditionally independent given the latent class H, therefore the shadows built from X_L and X_R commute. This inference is invalid. Theorem 3.3 requires conditional independence given the *meet*, not given some third variable.

Proposition 3.9 (Conditioning variable must be included)**.** *Let $X_L \perp X_R \mid H$ on a finite space with full support. Then:*

(a) $E_{\sigma(X_L,H)}E_{\sigma(X_R,H)} = E_{\sigma(H)} = E_{\sigma(X_R,H)}E_{\sigma(X_L,H)}$*, always;*

(b) $E_{\sigma(X_L)}$ *and* $E_{\sigma(X_R)}$ *commute iff* X_L *and* X_R *are conditionally independent given* $\sigma(X_L) \wedge \sigma(X_R)$*. With full support the meet is trivial, so this means* $X_L \perp X_R$ *marginally, which fails as soon as both channels depend on* H*.*

Proof. (a) is lemma 3.2 with $C = \sigma(H)$. (b) is theorem 3.3: with full support, $G(\sigma(X_L), \sigma(X_R))$ is complete bipartite, hence connected. □

Check A7 exhibits non-commuting $\sigma(X_L), \sigma(X_R)$ in random three-class instances and verifies (a) on all 200 of them. In the continuous Gaussian model of the previous version, the joint density of (X_L, X_R) is positive everywhere. So the meet is trivial, and $\sigma(Q_L(X_L))$ and $\sigma(Q_R(X_R))$ are not independent, because both are correlated with H. The claimed interchange for every ν is therefore false; its correct form is (a). The same lesson recurs for the tower shadows (section 4): a shadow must condition on the common centre.

4 Canonical shadows

4.1 Why label-defined measure-valued shadows do not compose

The previous version defined $P_A(\nu) = (m_A)_*\nu$ on $\mathsf{D}(\mathsf{D}X)$, with $m_A = \mathbb{E}_\nu[Z \mid \sigma(a)]$ for a label map $a\colon \mathsf{D}X \to I_A$. Composing two such maps applies the second label map to the *output* of the first.

Proposition 4.1 (Naive composites are single poolings)**.** *For label maps a_A, a_B and $\nu \in \mathsf{D}^2X$, let Z be the identity on* $(\operatorname{supp}\nu, \nu)$*. Then*

$$P_B(P_A(\nu)) = \operatorname{Law}\big(E_{A'}Z\big), \qquad A' = \sigma\big(a_B \circ m_A\big) \subseteq \sigma(a_A).$$

So $P_B \circ P_A$ is one pooling onto a coarsening of the first *partition. The "interchange" $P_BP_A = P_AP_B$ compares two different coarsenings, $\sigma(a_B \circ m_A)$ of $\sigma(a_A)$ and $\sigma(a_A \circ m_B)$ of $\sigma(a_B)$. It is not the operator identity $E_AE_B = E_BE_A$, and theorem 3.3 does not govern it.*

Proof. m_A is constant on $\sigma(a_A)$-cells, so $a_B \circ m_A$ is too. Pooling the atoms of $P_A(\nu)$ along a_B merges whole $\sigma(a_A)$-cells, and the barycentre of merged barycentres is the barycentre of the union (the tower property). □

Example 4.2 (Threshold labels)**.** On $\mathsf{D}\{0,1\} \cong [0,1]$, let ν be uniform on $\{0.1, 0.3, 0.7, 0.9\}$, with $a_A = [s \geq \frac{1}{2}]$ and $a_B = [s \geq \frac{1}{4}]$. Then $P_B(P_A\nu) = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}$, whereas $P_A(P_B\nu) = \frac{1}{4}\delta_{0.1} + \frac{3}{4}\delta_{19/30}$. On the fixed space the operators E_A, E_B also fail to commute, but with different values (check C3). So the two notions disagree even about *how* interchange fails.

Check C2 confirms proposition 4.1 on 200 random threshold pairs. The proposition also explains the uploaded scripts, which use nearest-atom labels (section 9).

4.2 Labelled random measures

Definition 4.3. A *labelled random measure* over X is a triple (I, p, Z): a finite probability space (I, p) and a map $Z\colon I \to \mathsf{D}X$. A morphism $f\colon (I, p, Z) \to (I', p', Z')$ is a map $f\colon I \to I'$ with $f_*p = p'$ and $Z' \circ f = Z$. The functor $\mathrm{Law}(I, p, Z) = Z_*p \in \mathsf{D}^2X$ forgets the labels.

Every $\Pi \in \mathsf{D}^2X$ is Law of the tautological object $(\operatorname{supp}\Pi, \Pi, \mathrm{id})$, but many labelled objects have the same law. A shadow is an operator on labelled objects, not on laws.

Proposition 4.4 (Shadows on a fixed space)**.** *For partitions A, B of I:*

(a) E_A is idempotent, preserves $\mathbb{E}Z$ (the barycentre of the law), and $\mathrm{Law}(E_AZ) \le_{\mathrm{cx}} \mathrm{Law}(Z)$;

(b) the composite of the shadows is the shadow of the composite: $\mathrm{Law}(E_B(E_AZ))$ is computed by the operator E_BE_A, so composition of shadows is *composition of conditional expectations;*

(c) interchange $\mathrm{Law}(E_AE_BZ) = \mathrm{Law}(E_BE_AZ)$ for all Z holds iff (i)–(v) of theorem 3.3 hold.

Proof. (a) and (b) are immediate on a fixed space. For (c), "if" follows from (i). For "only if" (with $|X| \ge 2$), the energy $\int \|v\|_2^2 \, d\,\mathrm{Law}(\cdot)$ is a function of the law. Write $Z = \rho_0 + Y$ with ρ_0 in the relative interior of Δ_X and Y small with zero coordinate sums. Conditional expectations fix constants and preserve means, so equality of energies gives $\|E_AE_BY\|^2 = \|E_BE_AY\|^2$ on an open set of Y, hence for all Y. By polarisation, $S := E_BE_AE_B = E_AE_BE_A$. The range of S lies in $L^2(A) \cap L^2(B) = L^2(M)$, $M = A \wedge B$, and S is self-adjoint, so $S = E_MSE_M = E_M$. For $Y \in L^2(B)$ this reads $\|E_AY\|^2 = \|E_MY\|^2$, so $E_AY = E_MY$ by Pythagoras. Hence $E_AE_B = E_M$, which is (ii). □

This discharges the "inheritance obligation" of the previous version (its Rem. 4.9): with shadows defined on a fixed labelled space, nothing needs to be inherited.

Proposition 4.5 (Naturality)**.** *Let $f\colon (I, p) \to (I', p')$ be measure preserving, and let $A = f^{-1}A'$, $B = f^{-1}B'$. Then for every Z',*

$$E_A(Z' \circ f) = (E_{A'}Z') \circ f, \qquad E_B(Z' \circ f) = (E_{B'}Z') \circ f.$$

So both shadows are natural with respect to the subcategory of measure-preserving maps that pull back both partitions.

Proof. For a cell a' of A', $f^{-1}(a')$ is a cell of A of mass $p'(a')$. The average of $Z' \circ f$ over it is the average of Z' over a'. □

This replaces the hypothetical "(L, R)-compatible subcategory" of the previous version (its Def. 5.10, Prop. 5.11) with a definite one, and makes its naturality a theorem.

4.3 The star space of a context

A coherent context specifies three laws that share the level-one predictive. Ω is a law over laws $\widetilde{\Pi}$ of predictives, Π is their mixture, and Σ couples labels θ with predictives. The context does not couple Ω with Σ. The canonical coupling is the conditional product over the shared predictive.

Definition 4.6 (Star space)**.** For a coherent context c, let $\kappa(\theta \mid r) = \Sigma(\theta, r)/\Pi(r)$. The *star space* of c is the finite probability space with points $(\widetilde{\Pi}, r, \theta, x)$ and weights

$$p_c(\widetilde{\Pi}, r, \theta, x) = \Omega(\widetilde{\Pi})\, \widetilde{\Pi}(r)\, \kappa(\theta \mid r)\, r(x),$$

restricted to positive weights. Write $\tilde{r}$ for the coordinate r, and $Z = \delta_x \in \mathsf{D}X$ for the observation, viewed as a vertex of the simplex.

Given $\tilde{r}$, the three *leaves* $\widetilde{\Pi}$, θ, x are mutually independent. Graphically the context is a star with centre $\tilde{r}$. By (1), the marginals are the context: $\mathrm{Law}(\widetilde{\Pi}) = \Omega$, $\mathrm{Law}(\tilde{r}) = \Pi$, $\mathrm{Law}(\theta, \tilde{r}) = \Sigma$ and $\mathrm{Law}(x) = \rho$ (check B1).

Definition 4.7 (Canonical shadows)**.** For a set $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}, \mathrm{obs}\}$ of leaves, let

$$G_T = \sigma(\tilde{r},\ \text{leaves in } T), \qquad S_T = E_{G_T}.$$

The *left shadow* is $L_c = S_{\{\mathrm{lev2}\}}$ (condition on the level-two law and the predictive). The *right shadow* is $R_c = S_{\{\mathrm{hyp}\}}$ (condition on the hypothesis label and the predictive).

Theorem 4.8 (The shadow algebra is a commutative band)**.** *For all leaf sets T, U,*

$$S_T S_U = S_{T \cap U} = S_U S_T.$$

Hence $T \mapsto S_T$ is a homomorphism from the meet-semilattice $(2^{\{\mathrm{lev2},\mathrm{hyp},\mathrm{obs}\}}, \cap)$ onto a commutative band of conditional expectations. In particular $L_c R_c = R_c L_c = S_\emptyset = E_{\sigma(\tilde{r})}$: the canonical shadows satisfy the interchange law strictly, and their common composite is the predictive shadow.

Proof. $G_{T\cap U} \subseteq G_T \cap G_U$. Given $(\tilde{r}, \text{leaves in } T \cap U)$, the leaves in $T \setminus U$ and those in $U \setminus T$ are independent, by mutual independence of the leaves given $\tilde{r}$. So G_T and G_U are conditionally independent given $G_{T\cap U}$, and lemma 3.2 applies. □

Check B2 verifies all 8×8 products exactly on 60 random coherent contexts with three-point base. The homomorphism need not be injective. For example, if θ is a function of $\tilde{r}$ (distinct labels per atom, as in the scripts' generator), then $S_{\{\mathrm{hyp}\}} = S_\emptyset$.

Proposition 4.9 (Dropping the centre)**.** *The label-only shadows $E_{\sigma(\widetilde{\Pi})}$ and $E_{\sigma(\theta)}$, which do not condition on $\tilde{r}$, commute iff the criterion of theorem 3.3 holds for the partitions $\sigma(\widetilde{\Pi})$ and $\sigma(\theta)$ of the star space. In random coherent contexts they fail to commute in 40 of 60 cases (check B3).*

The redefinition is therefore not cosmetic. The earlier left and right shadows (pooling by Π-atoms, pooling by Σ-hypotheses) are label-only shadows when read on a fixed space. They commute only in the special cases of theorem 3.3. The canonical shadows condition on the predictive $\tilde{r}$ that all levels share, and they always commute.

Proposition 4.10 (Markov absorption)**.** *For the observation $Z = \delta_x$:*

(a) $R_c Z = E_{\sigma(\tilde{r})} Z = \tilde{r}$, so the right shadow is invisible on observables;

(b) $E_{\sigma(\theta)} Z = m(\theta) := \sum_r \Sigma(r \mid \theta)\, r$, and $E_{\sigma(\tilde{r})} E_{\sigma(\theta)} Z \neq E_{\sigma(\theta)} Z$ in general (59 of 60 random contexts, check B5).

Proof. (a) $x \perp \theta \mid \tilde{r}$. (b) is a direct computation; the inequality holds whenever $m(\theta)$ is not a function of $\tilde{r}$. □

So the hypothesis level is bookkeeping: it does not move the prediction given the predictive. It moves the prediction only once the predictive is forgotten, and then in an order-dependent way.

5 Algebra II and topology: how the tower acts

5.1 Two idempotents

The monad unit laws $\mu \circ \eta_{\mathsf{D}} = \mathrm{id} = \mu \circ \mathsf{D}(\eta)$ give one retraction $\mu\colon \mathsf{D}^2X \to \mathsf{D}X$ with two sections. Hence there are two split idempotents on D^2X:

$$e_\perp = \eta_{\mathsf{D}X} \circ \mu \ (\Pi \mapsto \delta_{\mu\Pi}), \qquad e_\top = \mathsf{D}(\eta_X) \circ \mu \ (\Pi \mapsto \textstyle\sum_x (\mu\Pi)(x)\, \delta_{\delta_x}).$$

Proposition 5.1 (Two idempotents)**.** *(a) $e_\perp$ and $e_\top$ are idempotent, natural in X, and split through the same retraction μ.*

(b) They form a left-zero band: $e_\perp e_\top = e_\perp$ and $e_\top e_\perp = e_\top$. So $\{\mathrm{id}, e_\perp, e_\top\}$ is closed under composition.

(c) $e_\top$ is affine and $e_\perp$ is not (79 of 80 random mixtures, check B6b).

(d) For the W_1 metric on $\mathsf{D}(\Delta_X)$ built from ℓ^1: μ is 1-Lipschitz, η and $\mathsf{D}(\eta)$ are isometries, and $e_\perp, e_\top$ are 1-Lipschitz, hence continuous for the weak topology.

Proof. (a) Idempotence and the splitting are the unit laws. Naturality follows from naturality of η and μ. (b) $e_\perp e_\top = \eta\mu\mathsf{D}(\eta)\mu = \eta\mu = e_\perp$, and $e_\top e_\perp = \mathsf{D}(\eta)\mu\eta\mu = \mathsf{D}(\eta)\mu = e_\top$. (c) μ and $\mathsf{D}(\eta)$ are affine (pushforward is linear in the measure). η is not: $\delta_{s\rho+(1-s)\rho'} \neq s\delta_\rho + (1-s)\delta_{\rho'}$. (d) For $g \in [-1,1]^X$, $r \mapsto \langle g, r\rangle$ is 1-Lipschitz for ℓ^1, so $\|\mu\Pi - \mu\Pi'\|_1 \le W_1(\Pi, \Pi')$. Also $W_1(\delta_\rho, \delta_{\rho'}) = \|\rho - \rho'\|_1$. Distinct vertices are at distance 2, so $W_1(\mathsf{D}(\eta)\rho, \mathsf{D}(\eta)\rho') = 2\,\mathrm{TV}(\rho, \rho') = \|\rho - \rho'\|_1$. The two idempotents are compositions of these maps. □

The tower paper's pooling idempotent is $e = e_\perp$ [1, §13.6], [2]. Its partner $e_\top$ is what the unit of C_3 produces at level one (proposition 5.4).

5.2 Barycentre fibres

Proposition 5.2 (Fibres are convex-order intervals)**.** *For $\rho \in \mathsf{D}X$, let $F_\rho = \mu^{-1}(\rho)$ be the set of Borel probability measures on the simplex with barycentre ρ (in this section $\mathsf{D}(\Delta_X)$ denotes all Borel probability measures on Δ_X, of which the finitely supported ones are a dense subset). Then:*

(a) F_ρ is convex and weakly compact, and its finitely supported elements are weakly dense in it;

(b) for every $\Pi \in F_\rho$, $e_\perp\Pi = \delta_\rho \ \le_{\mathrm{cx}}\ \Pi \ \le_{\mathrm{cx}}\ \mathsf{D}(\eta)\rho = e_\top\Pi$;

(c) $\mathsf{D}(\eta)\rho$ is the unique element of F_ρ supported on the vertices.

So F_ρ is a convex-order interval $[e_\perp\Pi, e_\top\Pi]$, the same for every $\Pi \in F_\rho$. The two idempotents are the retractions of the fibre onto its bottom and its top.

Proof. (a) Δ_X is compact metric, so $\mathsf{D}(\Delta_X)$ is weakly compact (Prokhorov), and μ is continuous by proposition 5.1(d), so F_ρ is closed; it is convex because μ is affine. For density, partition Δ_X into finitely many Borel cells of diameter $< \varepsilon$ and replace Π on each cell by a point mass at the cell's conditional barycentre. This is a pooling, so it keeps the barycentre ρ, and it moves Π by less than ε in W_1. (b) The coupling $\rho \mapsto r$ with law Π is a martingale coupling of δ_ρ and Π. The coupling $r \mapsto \delta_x$ with probability $r(x)$ is a martingale coupling of Π and $\mathsf{D}(\eta)\rho$ (check B6). (c) Vertices are affinely independent, so a measure $\sum_x w_x \delta_{\delta_x}$ with barycentre ρ has $w = \rho$. This is the Choquet–Meyer uniqueness of maximal representing measures on a simplex [10]. □

5.3 A coherent context is a martingale

Theorem 5.3 (The tower as a martingale)**.** *On the star space of a coherent context, let $\mathcal{F}_0$ be trivial, $\mathcal{F}_1 = \sigma(\widetilde{\Pi})$, $\mathcal{F}_2 = \sigma(\widetilde{\Pi}, \tilde{r})$ and $\mathcal{F}_3 = \sigma(\widetilde{\Pi}, \tilde{r}, x)$, and put $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$ with $Z = \delta_x$. Then $(M_k)_{k=0}^3$ is a Δ_X-valued martingale with*

$$\mathrm{Law}(M_0) = \delta_\rho, \quad \mathrm{Law}(M_1) = \mathsf{D}(\mu)(\Omega), \quad \mathrm{Law}(M_2) = \Pi, \quad \mathrm{Law}(M_3) = \mathsf{D}(\eta)(\rho),$$

so that

$$\delta_\rho \ \leq_{\mathrm{cx}} \ \mathsf{D}(\mu)\Omega \ \leq_{\mathrm{cx}} \ \Pi \ \leq_{\mathrm{cx}} \ \mathsf{D}(\eta)\rho$$

is a chain in the fibre F_ρ. The hypothesis branch $\sigma(\theta, \tilde{r})$ joins the chain at $\mathcal{F}_2$ without moving it: $\mathbb{E}[Z \mid \theta, \tilde{r}] = M_2$ (proposition 4.10(a)).

Proof. $M_0 = \mathbb{E}Z = \rho$. $M_1 = \sum_r \widetilde{\Pi}(r)\, r = \mu(\widetilde{\Pi})$, which has law $\mathsf{D}(\mu)\Omega$. $M_2 = \tilde{r}$, which has law Π. $M_3 = \delta_x$, which has law $\mathsf{D}(\eta)\rho$. The chain follows from proposition 5.2(b) and Strassen's theorem (check B4). □

This is the structural meaning of the tower: its levels are the successive conditional expectations of the observation along one filtration, and coherence (1) is the martingale property in law. A shadow that conditions on an initial segment of the filtration lands on a level, and nested shadows commute (theorem 3.3). The shadow algebra of theorem 4.8 adds the hypothesis leaf to this chain at the level of the predictive.

5.4 The Beck defects located

Proposition 5.4 (Where the defects of λ^σ sit)**.** *At level one:*

(a) the two sides of (U) for λ^σ at (x, c) are $e_\perp\Pi$ and $e_\top\Pi$ for any $\Pi \in F_\rho$, namely δ_ρ and $\mathsf{D}(\eta)\rho$. So the unit defect is the full convex-order diameter of the fibre F_ρ;

(b) for $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k}\delta_{(x_{\ell k}, c_{\ell k})}$, take the sample-index space $I = \{(\ell, k)\}$ with $p(\ell, k) = \nu_\ell a_{\ell k}$ and $W(\ell, k) = \rho_{\ell k}$. Then the two sides of (M) are $\mathrm{Law}(W)$ (λ-first) and $\mathrm{Law}(\mathbb{E}[W \mid \ell])$. So the multiplication defect is one conditional-expectation step, onto the group label.

Proof. (a) By [1, Prop. 13.8(a)], the left side is $\delta_\rho = e_\perp(\mathsf{D}(\eta)\rho)$ and the right side is $\mathsf{D}(\eta)\rho = e_\top(\delta_\rho)$. (b) By [1, Prop. 13.8(b)], the right side is $\sum_\ell \nu_\ell \delta_{\bar{\rho}_\ell}$ with $\bar{\rho}_\ell = \sum_k a_{\ell k}\rho_{\ell k} = \mathbb{E}[W \mid \ell]$. Both statements are verified with the tower code's own λ^σ, C_3, η and μ (check B7). □

In this language the orientation obstruction of [1, Prop. 13.9, Thm 13.15] becomes visible. In (U) the λ-first side is the bottom of a fibre and the other side its top; in (M) the λ-first side is the unpooled law and the other side a conditional expectation of it. The shift law *pools after the unit and spreads after the multiplication.* Every order that reads conditional expectation monotonically therefore orients the two defects oppositely. [1, Prop. 13.17] shows that one instance of (U) is an instance of (M) read backwards.

5.5 Continuity: canonical versus label-defined shadows

Proposition 5.5. *(a) On a fixed finite index with fixed partitions, $(p, Z) \mapsto \mathrm{Law}(E_A Z)$ is continuous on full-support weights, and $W_1(\mathrm{Law}(E_A Z), \mathrm{Law}(E_A Z')) \leq \|Z - Z'\|_{L^1(p)}$.*

(b) *Nearest-atom shadows are Borel but not continuous. In the configuration of check C4 (atoms* $(\frac{2}{3}, \frac{1}{3})$ *and* $(\frac{1}{3}, \frac{2}{3})$ *on a two-point base), two inputs at* W_1*-distance* $\frac{1}{1000}$ *have outputs at* W_1*-distance* $\frac{501}{2000}$.

Proof. (a) Cell averages are continuous in (p, Z) when cell masses are positive, and conditional expectation is an L^1 contraction. (b) A point crossing the Voronoi wall between the atoms changes cells. The cell barycentres then jump by a fixed amount (exact computation, check C4). □

So the topology separates the two definitions. Canonical shadows vary continuously with the context. Nearest-atom shadows have walls, and any statement about them on finitely many random instances depends on which side of the walls the instances fall. Ties on the walls are also broken by key order, which explains several artefacts in section 9.

6 Two-dimensional structure: what is and is not a double category

6.1 Why the previous double-category theorem is withdrawn

The previous version's Def. 11.1 took 2-cells to be "natural transformations $\alpha\colon g \circ L \Rightarrow R \circ f$". But L and R are functions between sets, not functors. In the locally discrete 2-category **Meas**, the only 2-cells are identities [1, §§13.1, 13.5], so such an α exists only when $g \circ L = R \circ f$. The theorem also cited definitions (6.1.1, 6.2.1, 6.3.1) that do not exist, and its own Prop. 6.1 lists data that were never supplied. Theorems 11.2 and 11.5 are therefore withdrawn. What survives is the following correct and fairly general mechanism.

6.2 Quintets of a locally preordered 2-category

Proposition 6.1. *Let* $\mathcal{K}$ *be a 2-category whose hom-categories are preorders. Its quintet double category* $Q(\mathcal{K})$ *is defined as follows:*

- *objects and both kinds of arrows are those of* $\mathcal{K}$*;*
- *a square with boundary* $(h\colon a \to b,\ v\colon a \to c,\ w\colon b \to d,\ k\colon c \to d)$ *exists, uniquely, iff* $k \circ v \leq w \circ h$.

Then $Q(\mathcal{K})$ *is a strict double category, and interchange, associativity and unit laws for squares hold automatically.*

Proof. This is Ehresmann's quintet construction [5, 7]. Pasting composes inequalities by transitivity and by whiskering (monotonicity of composition), and uniqueness of 2-cells makes every equation between squares trivial. □

This is the honest content of "the interchange law is automatic". It holds once the 2-category is locally preordered and composition is monotone. Everything substantive lies in choosing $\mathcal{K}$.

6.3 What the tower supplies, and what it lacks

(a) *Convex order on one level.* On $\mathsf{D}(K)$ for convex K, $\leq_{\mathrm{cx}}$ is a partial order. Pushforward along affine maps is monotone [1, §13.6]. The shadows of theorem 4.8 give a monotone map $T \mapsto \mathrm{Law}(S_T Z)$ from the Boolean lattice of leaf sets into $(F_\rho, \leq_{\mathrm{cx}})$, and the chain of theorem 5.3 is a chain of vertical arrows.

(b) *Strict interchange for canonical shadows.* By theorem 4.8, the square with vertical sides L_c, R_c and corner $S_\emptyset$ commutes strictly.

(c) *Lax interchange in general.* By corollary 3.8, two non-commuting shadows give a cospan of vertical arrows into the meet. This is a pair of squares in $Q(\mathcal{K})$, not an equality.

(d) *The Beck defects.* The (U) and (M) defects of λ^σ are vertical arrows of opposite orientation relative to their λ-first sides (proposition 5.4; [1, Thm 13.15]). So λ^σ is neither a lax nor an oplax morphism in any such $Q(\mathcal{K})$ whose order reads conditional expectation monotonically. The (Cm) defect of $\bar{\lambda}$ is incomparable in the joint convex order [1, Prop. 13.10] but is a cell of one orientation for label pooling at every level [1, Thm 13.16].

(e) *What is missing.* The functor D must act monotonically on hom-preorders. The pointwise convex order does not suffice: $\delta_a \leq_{\mathrm{cx}} \delta_b$ forces $a = b$, so $\mathsf{D}(f) \leq \mathsf{D}(g)$ fails even when $f \leq g$ pointwise. The natural candidate is the relation (Barr) lifting of the preorder along D. Since D preserves weak pullbacks, this lifting preserves reflexivity and transitivity and makes D monotone. Whether C_3, the structure maps and $\bar{\lambda}$'s label-pooling cells are monotone for the lifted order is the tower paper's open problem 4, now with a definite candidate.

6.4 Options A and B

Proposition 6.2. *Let $\mathsf{C}_E Y = Y \times \prod_i E_i Y$ be built from functors E_i, and let λ be assembled from component transports $\lambda_i \colon \mathsf{D}E_i \Rightarrow E_i\mathsf{D}$. Then (U), (Co) and (M) for λ hold iff they hold for every component. (Cm) does not decompose this way, because the comultiplication $\delta(y, c) = ((y, c), \prod_i E_i(j_c)(c_i))$ feeds the* whole *context c into every component through j_c.*

Proof. The unit, counit and multiplication of D act on each factor separately, and equality in a product is componentwise. The formula for δ is [1, §6]. □

So the previous Thms 7.1 and 7.2 ("options A and B satisfy all four axioms if the components do") are valid for (U), (Co) and (M) only. For (Cm), [1, Thm 13.16] shows that the averaging-type transport has its defect in every component at once.

7 The Giry extension

Let X be Polish and let $\mathcal{P}(X)$ carry the weak topology. Then $\mathcal{P}$ is a monad on Polish spaces, with continuous η and μ [6], and on finite X it is D. The constructions above extend as follows.

(i) *Shadows.* A labelled random measure is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a measurable $Z \colon \Omega \to \mathcal{P}(X)$. For $G \subseteq \mathcal{F}$, $\mathbb{E}[Z \mid G](B) := \mathbb{E}[Z(B) \mid G]$ has a version that is a random probability measure when X is standard Borel (regular conditional distributions). Shadows are $Z \mapsto \mathbb{E}[Z \mid G]$, and propositions 4.4 and 4.5 hold verbatim.

(ii) *Commutation.* (i)⇔(ii)⇔(iii) of theorem 3.3 hold for sub-σ-algebras of any probability space, applied to $Z(B)$ for B in a countable generating π-system; the proof given is general. The finite criteria (iv) and (v) have no direct analogue: the meet is no longer the set of components of a graph, and $A \cap B$ need not be countably generated. They are replaced by the spectral condition $E_A E_B E_A = E_{A \wedge B}$ on $L^2(A)$.

(iii) *Rate.* theorem 3.6 holds with the Friedrichs cosine c. When $L^2(A) + L^2(B)$ is not closed, $c = 1$ and the lax interchange of corollary 3.8 holds only in the strong limit, with no rate.

(iv) *Fibres and martingale.* proposition 5.2(a,b) and theorem 5.3 hold for X compact metric, with $\mathsf{D}(\eta)\rho$ replaced by $\mathcal{P}(\eta)\rho$. For (c), $\mathcal{P}(X)$ is a Bauer simplex, so maximal representing measures are unique [10].

Open problem 7.1. In the infinite case with $c = 1$, does the lax interchange cospan of corollary 3.8 admit a coherent choice, natural along measure-preserving maps that do not pull back the σ-algebras? In finite dimensions it is canonical; in general it exists only as a strong limit.

8 Analytic and algebraic outlook, corrected

8.1 The finite stochastic-to-convex functor

The previous version's Theorem 14.3 is correct, and it can be sharpened: $T\colon \mathrm{Kl}(\mathsf{D})_{\mathrm{fin}} \to \mathbf{Conv}$, $K \mapsto T_K$, is *fully faithful*. An affine map $\Delta_X \to \Delta_Y$ is determined by the images of the vertices, which may be any points of Δ_Y, that is, any stochastic matrix (check E1). This is the comparison functor from the Kleisli category to the Eilenberg–Moore category, restricted to free algebras. Two statements of its §14.6 are wrong. The shadows are not affine (proposition 5.1(c)), so they are not in the image of T. And $\mathsf{D}(\Delta_X)$ is not a finite simplex; the iterated levels are infinite-dimensional convex sets.

8.2 The Berkovich construction

Its Def. 15.3 sets $A_X = C(\Delta_X) \otimes_{\mathbb{R}} k$ for a complete non-Archimedean field k and calls this affinoid. This construction has three problems.

(a) There is no isometric embedding $\mathbb{R} \hookrightarrow k$: the absolute value of $\mathbb{R}$ is Archimedean, so a tensor norm compatible with both is not defined.

(b) $C(\Delta_X)$ is not a quotient of a Tate algebra, and the tensor product would not be affinoid.

(c) Over $\mathbb{R}$, Gel'fand duality gives $\mathcal{M}(C(\Delta_X)) \cong \Delta_X$. The map $K^*(f) = f \circ T_K$ is a unital algebra homomorphism, and $\mathcal{M}(K^*) = T_K$. So the real construction returns T and adds nothing.

Its Theorem 15.4 and Proposition 15.7 are therefore withdrawn as statements about Berkovich spaces; (c) is their correct real content. A genuine analytic route exists through tropical geometry. The log-linear models of the conditionals paper [3] are toric, their tropicalisations are polyhedral fans, and Payne's theorem [9] identifies the Berkovich analytification of a toric embedding with the inverse limit of its tropicalisations. Relating the tower's barycentre fibres to these skeleta is open.

8.3 Moment varieties

Proposition 8.1. *(a) The order-r moment map of a two-component univariate Gaussian mixture has Jacobian rank* $\min(r, 5)$ *at a generic point (check D1 at a rational point for* $r = 2, \dots, 6$*). At order two the map fills* $\mathbb{R}^2$*. Mixtures are invisible to second moments, and local identifiability needs order at least five.*

(b) Every affine statistic of a flattened law is constant on barycentre fibres. Regrouping a mixture of mixtures does not change its moments (check D2). So the moment image of any tower level equals that of its barycentre $\mu^{(k)}$*.*

Proof. (a) Direct computation; the order-two image $\{(m, v + m^2)\}$ of a single Gaussian is already open. (b) Moments are integrals, hence affine in the measure, and μ preserves integrals of affine functions. □

Hence the previous version's §16 needs two changes. The moment variety must have order at least three (univariate: at least five for two components). And secant varieties see only μ: the hierarchical information of the tower lives in the convex-order fibre (proposition 5.2), not in the moment image. The "walls" of a component collision are the points of Sec_{k-1} inside Sec_k, where the k-component parametrisation has positive-dimensional fibres (the weight split). Defectivity is a different notion: it is a global property of the whole secant variety, not a locus. The chain $\mathrm{Sec}_1 \subseteq \mathrm{Sec}_4 \subseteq \mathrm{Sec}_{16}$ is specific to 4-ary trees; in general it is Sec_{m^n}. The proposed wall-crossing formula remains a proposal.

9 Computational checks and the audit of the earlier scripts

9.1 The new checks

`check_shadows.py` (in this folder) runs 26 checks, all in exact rational arithmetic except the spectral checks A1(v) and A4, which use double precision with tolerance 10^{-9}. Tower objects are imported verbatim from `check_tower.py`. All 26 pass (results hash in `FROZEN.txt`). Table 2 maps them to statements.

9.2 Audit of the five scripts of the previous version

All five scripts share one `pooling(nu, atoms)`: assign each support point to the nearest atom in a "type-aware" total variation, break ties by key order, and replace each group by its barycentre. Running them reproduces their reported counts (every check passes in every trial). Table 3 explains why each count says nothing about the theorem it was meant to test.

Table 3: What the earlier scripts measure.

Script	Reports	Why it passes
`check_giry.py`	interchange, idempotence, barycentre, 3-cell, "disintegration" 300/300	The generator gives Π exactly two atoms, and Σ's descriptors i mod 2 biject with them. So $\sigma(\theta) = \sigma(\tilde{r})$ (committed regime) and L and R have identical cells (C1). The "disintegration" test only checks that barycentres sum to one.
`check_3cat-Coherence.py`	options A/B interchange, coherence, 3-cell	As above. In addition, the level-5 atoms δ_ρ live in D^2 while the inputs live in D^1, so every TV distance is 1 (disjoint keys), all points tie, and the pooling is the total pooling $e_\perp$ (C7). $e_\perp$ commutes with every barycentre-preserving map, so the "3-cell" is trivial for that reason.

Script	Reports	Why it passes
`check_noncoherent.py`	$L \neq R$ in 0/300; interchange 300/300	The duplicated predictive $(0, \rho_1), (1, \rho_1)$ ties, and the first key wins. $(1, \rho_1)$ never receives mass, so R's cells equal L's (C6). The "non-coherent" case is coherent in effect.
`double_category.py`	interchange 300/300; (U) defect nonzero	The squares never use the shift. "Interchange" is $R(L\nu) = L(R\nu)$ with $L = R$. "(U) defect nonzero" is $\delta_\rho \neq \mathsf{D}(\eta)\rho$, i.e. $e_\perp \neq e_\top$ (proposition 5.4), which is true but is not a double-category statement.
`check_span.py`	$L \neq R$ in 0/300 for every input type; all else 300/300	This is why it "did not work". (1) The cells coincide (C1). (2) `dist_of` strips θ, so hypothesis inputs become distribution inputs after one step, and the outputs' keys have a different type. "L not inverse of ν" therefore holds automatically (C5). (3) "(Cm) defect equals sum" compares (L, R) with a recomputation of (L, R), a tautology (C5). (4) Mixed inputs combine keys of two types under one metric.

How to make the span test meaningful. Replace nearest-atom pooling with conditional expectation on the star space (definition 4.6). Draw Σ with labels that are *not* functions of the predictive. Then compare three pairs: the canonical shadows L_c, R_c (always commute, theorem 4.8); the label-only shadows $E_{\sigma(\widetilde{\Pi})}, E_{\sigma(\theta)}$ (commute iff theorem 3.3); and the naive measure-valued composites (proposition 4.1). Checks B2, B3 and C2 do exactly this, and they separate the three behaviours.

10 Remark added to the tower paper

The tower paper is correct as it stands: it never uses $P_B P_A = P_{A \vee B}$. Its [1, Def. 13.13] takes poolings on a fixed sample index, and [1, Thm 13.16] defines its order as a transitive closure. One unnumbered remark has been added after Thm 13.16 to say why the closure is needed. A two-step label pooling $E_{G_2} E_{G_1}$ is itself a single label pooling, i.e. a conditional expectation, iff E_{G_1} and E_{G_2} commute, which by theorem 3.3 means conditional independence given $G_1 \wedge G_2$. In general it is not, and the closure is genuinely larger than the one-step relation.

11 Summary and open problems

- The interchange question for two conditional expectations is completely answered in finite dimensions. They commute iff they are conditionally independent given their meet, which is checkable as rank-one blocks on graph components or as singular values in $\{0, 1\}$. In general they interchange laxly, through the meet, at rate c^{2n-1}. The join identity is false except for equal σ-algebras.

- Shadows must be composed on a fixed probability space. A coherent context supplies one canonically: the star space centred at the level-one predictive. There the canonical shadows form a commutative band and interchange strictly. The earlier label-only shadows do not, unless the spectral criterion holds.
- The tower acts on barycentre fibres. Its levels are the positions of one martingale; $e_\perp$ and $e_\top$ are the bottom and top retractions; the unit defect of λ^σ is the full diameter of a fibre, and its multiplication defect is one conditional-expectation step.

Open problems.

1. Open problem 7.1: coherence of the lax interchange in infinite dimensions when $c = 1$.
2. A 2-category for the tower: is the Barr lifting of label pooling along D preserved by C_3 and its structure maps, so that $\bar{\lambda}$ becomes a lax law (tower open problem 4)?
3. Higher-level fibres: describe the convex-order structure of $(\mu^{(k)})^{-1}(\rho)$ for $k = 2, 3$, and locate the nested pooling idempotents P_k of [2] in it.
4. Tropical skeleta: relate barycentre fibres of toric models to the skeleta of their analytifications [9].

Code and data. `check_shadows.py` and `FROZEN.txt` accompany this paper. No data are used.

References

[1] V. K. Awasthi, The three-level posterior tower: hierarchical Bayesian inference and mixed distributive laws. Technical report, October 2026 (version 10).

[2] V. K. Awasthi, Pooling idempotents and the Beck defects of the absolute tower. Note, October 2026.

[3] V. K. Awasthi, Conditionals, couplings and toric geometry. Technical report, October 2026.

[4] H. H. Bauschke, F. Deutsch and H. Hundal, Characterizing arbitrarily slow convergence in the method of alternating projections. *Int. Trans. Oper. Res.* **16** (2009), 413–425.

[5] C. Ehresmann, Catégories structurées. *Ann. Sci. Éc. Norm. Sup.* **80** (1963), 349–426.

[6] M. Giry, A categorical approach to probability theory. In *Categorical Aspects of Topology and Analysis*, LNM 915, Springer, 1982, 68–85.

[7] M. Grandis, *Higher Dimensional Categories: From Double to Multiple Categories*. World Scientific, 2019.

[8] S. Kayalar and H. L. Weinert, Error bounds for the method of alternating projections. *Math. Control Signals Systems* **1** (1988), 43–59.

[9] S. Payne, Analytification is the limit of all tropicalizations. *Math. Res. Lett.* **16** (2009), 543–556.

[10] R. R. Phelps, *Lectures on Choquet's Theorem*, 2nd ed. LNM 1757, Springer, 2001.

[11] V. Strassen, The existence of probability measures with given marginals. *Ann. Math. Statist.* **36** (1965), 423–439.

[12] J. von Neumann, *Functional Operators, Vol. II*. Princeton University Press, 1950.

Table 1: Claims of the previous version and their status in this revision.

Previous claim	Status	Replacement
$P_B P_A = P_{A \vee B}$ (used for interchange)	false	holds iff $A = B$ (proposition 3.4); the correct product is $E_{A \wedge B}$, under conditional independence given the meet (theorem 3.3)
Thm 4.4: conditional independence $\Rightarrow$ commute	true, one-sided	two-sided equivalence with three further conditions (theorem 3.3); nestedness is a special case
Rem 4.7 "without the hypotheses the law fails"	imprecise	the hypotheses are necessary and sufficient for the operators; in general a lax interchange with exact rate (theorem 3.6 and corollary 3.8)
Def 3.2 shadows via label maps on $\mathsf{D}(B)$	ill-suited	composites re-partition (proposition 4.1); shadows are redefined on the star space of a context (definitions 4.6 and 4.7)
Prop 3.3 idempotence under label preservation	correct but unnecessary	idempotence is automatic on a fixed space
Prop 4.8, 8.1 "inheritance"	obligation	discharged: on a fixed space composition of shadows *is* composition of conditional expectations (proposition 4.4)
Prop 5.11 naturality, hypothesis-dependent	conditional	theorem for measure-preserving maps that pull back the σ-algebras (proposition 4.5)
Thm 11.2, 11.5 double category of the tower	not established	2-cells mistyped; replaced by the quintet construction for locally preordered 2-categories and an exact statement of what it does and does not give (section 6)
Prop 11.7 Gaussian interchange for every ν	false	conditional independence given H does not make $\sigma(X_L)$, $\sigma(X_R)$ commute; with H included it does (proposition 3.9)
Thms 7.1, 7.2 options A and B "componentwise"	partly invalid	(U), (Co), (M) are componentwise, (Cm) is not (proposition 6.2)
§14.6 shadows are affine maps	false	$e_\perp$ and the shadows are not affine (proposition 5.1)
§15 functor to Berkovich spaces	ill-defined	$\mathbb{R}$ does not embed isometrically in a non-Archimedean field; over $\mathbb{R}$ the construction returns T (section 8)
§16 Gaussian moment variety of order two	incorrect	order-two moments fill the ambient space; mixtures need order ≥ 3 (proposition 8.1)
3-cell triviality and interchange in the scripts	artefacts	audit in section 9

Table 2: Checks in `check_shadows.py`.

A1–A2	theorem 3.3 five-way equivalence (449 commuting, 151 non-commuting instances); proposition 3.4 (join identity held in 145 of 400 trials, all with $A = B$)
A3–A5	example 3.5; theorem 3.6 for $n \leq 6$ on 157 pairs; corollary 3.8 exactly on 300 pairs
A6–A7b	lemma 3.2; proposition 3.9
B1–B5	star space coherence; theorem 4.8 (all 8×8 products, 60 contexts); proposition 4.9; theorem 5.3; proposition 4.10
B6–B7	propositions 5.1 and 5.2; proposition 5.4 with the tower's own λ^σ
C1–C7	audit of the uploaded scripts (below); proposition 4.1; example 4.2; proposition 5.5(b)
D1–E1	proposition 8.1; full faithfulness of T